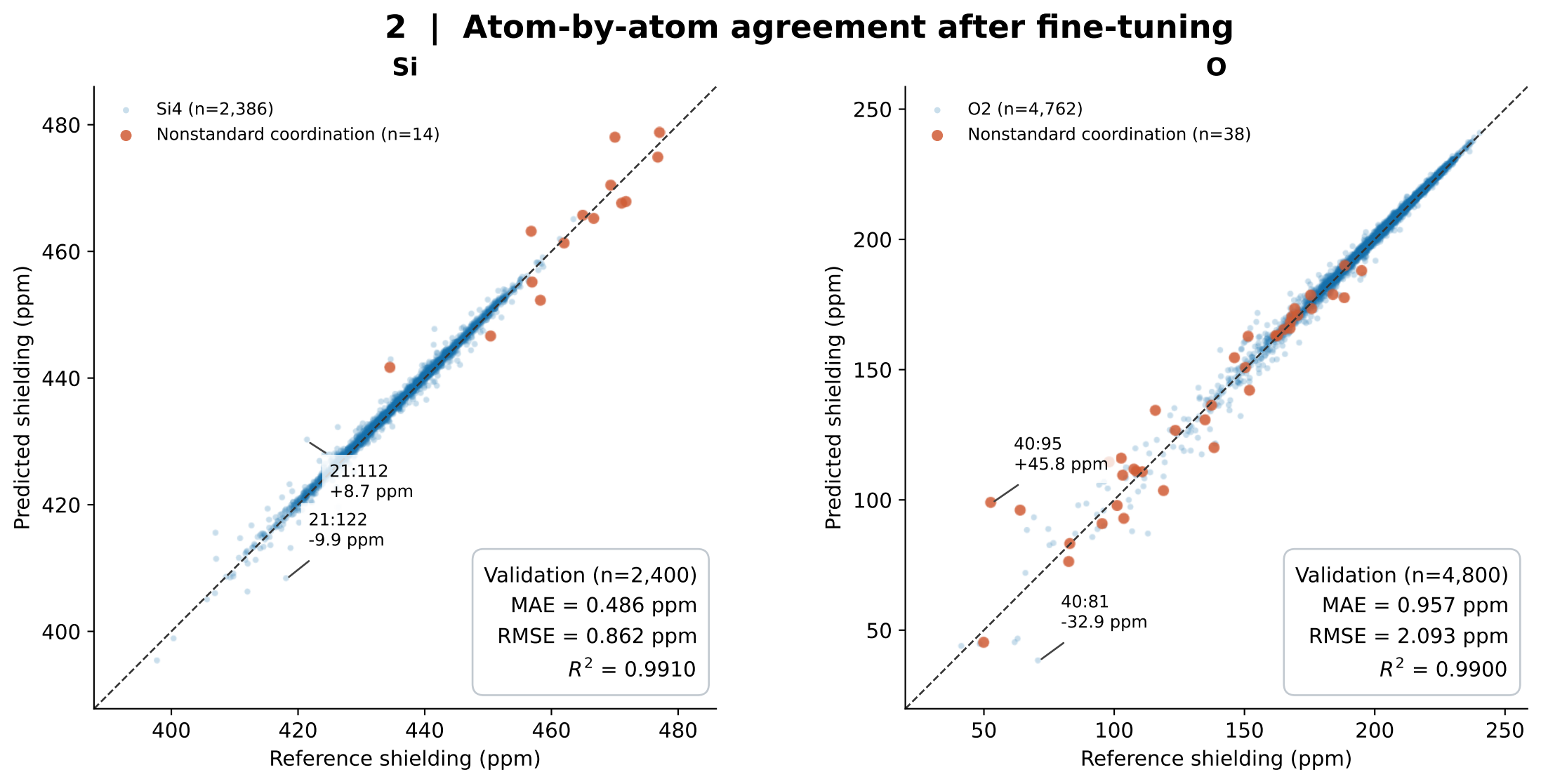

2 | Atom-by-atom agreement after fine-tuning
Si
Si4 (n=2,386)
Nonstandard coordination (n=14)
21:112
+8.7 ppm
21:122
-9.9 ppm
Validation (n=2,400)
MAE = 0.486 ppm
RMSE = 0.862 ppm
R^2 = 0.9910
Predicted shielding (ppm)
Reference shielding (ppm)
400
420
440
460
480
O
O2 (n=4,762)
Nonstandard coordination (n=38)
40:95
+45.8 ppm
40:81
-32.9 ppm
Validation (n=4,800)
MAE = 0.957 ppm
RMSE = 2.093 ppm
R^2 = 0.9900
50
100
150
200
250